

Experimental Data Acquisition Protocol
Single-LED / Single-PD 3D OWP Validation

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1. Objective

Acquire a structured and reproducible experimental dataset to validate 3D optical positioning using a single LED transmitter and a single PD under controlled receiver reorientation. The same acquisition sequence will be applied at every spatial position.

2. Acquisition design

Parameter	Current definition	Notes
XY measurement area	2 m x 2 m	Regular Cartesian grid
XY spacing	0.25 m	9 x 9 = 81 XY points per z level
Z sampling	TBD	Multiple z levels; range and spacing to be defined before acquisition
Number of orientations	K = 5	Same orientation set at every spatial point
Azimuth angles	{0°, 72°, 144°, 216°, 288°}	Uniformly spaced over 360°
Tilt angle	15°	Fixed tilt for all orientations

3. Experimental setup and reference frame

Transmitter: LED fixed at a known reference location.

Receiver: Photodetector mounted on the MTRS, with gantry-controlled x, y, and z motion.

Acquisition: Keysight MSOX6004A oscilloscope.

Reference frame: The coordinate origin, axis directions, LED position, receiver reference point, and azimuth/tilt conventions will be documented before acquisition.

4. Acquisition procedure

- S0.** Record a background measurement before starting the acquisition campaign.
- S1.** For each selected z level, sequentially move the receiver through the 2 m × 2 m XY grid with 0.25m spacing.
- S2.** At every reference position (x, y, z), sequentially command the same K = 5 receiver orientations and allow the motion stage to settle.
- S3.** For each orientation, acquire and store the oscilloscope waveform together with the commanded position, orientation, and acquisition settings.

5. Data record and traceability

Each measurement record will include: measurement ID; ground-truth (x, y, z); commanded/recorded MTRS orientation; raw oscilloscope data; extracted signal metric(s); LED operating parameters; oscilloscope settings; timestamp; and relevant setup notes.